

YUDHISTIRA ONGGOWARSITO

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EXECUTIVE SUMMARY

- AI-focused Software Engineer with 3+ years at Traveloka building automation and backend systems on AWS (Java, Spring Boot, Python), currently specialising in machine learning.
- Hands-on experience with the modern ML/AI stack: PyTorch, Hugging Face Transformers, sentence-transformer embeddings, recommender systems (RecBole, SASRec, EASE), and LLM-based agentic and RAG systems (LangGraph, ChromaDB).
- Proven delivery across contexts: leading a 4-engineer team on an agentic AI product, independent freelance automation for clients, and enterprise workflow automation saving ~500 man-hours annually at Traveloka.

SKILLS

- **Programming:** Python, Java, SQL, NoSQL, JavaScript
- **ML / AI:** PyTorch, scikit-learn, Hugging Face Transformers, Sentence-Transformers, RecBole, Pandas, NumPy; recommender systems (SASRec, UniSRec, EASE/ELSA), RAG, LLM agents (LangGraph), prompt engineering, vector databases (ChromaDB), embeddings, active learning, XGBoost, Graph Neural Networks
- **Cloud & Infrastructure:** AWS (Lambda, ECS, EC2, S3, DynamoDB, DocumentDB, SQS, SNS, Athena, CloudWatch, Route 53, ELB, IAM, KMS, Cognito), GCP, Docker, Terraform, CI/CD (GitHub Actions), microservices, REST/OpenAPI, Airflow, Grafana, Spring Boot, Google Apps Script
- **AI-Assisted Development:** GitHub Copilot in production development at Traveloka, Claude and Gemini coding agents for agentic coding, code review, and rapid prototyping across SMU coursework and personal projects
- **Certification:** AWS Certified Developer – Associate
- **Languages:** English, Bahasa Indonesia, Mandarin

PROFESSIONAL EXPERIENCE

INDIRATEX SPINDO — Indonesia

Mar 2026 – Present

Technical Advisor

- Lead a team of 4 engineers driving a manufacturing company's digital transformation, owning delivery of an agentic AI product that streamlines internal operations and reduces process delays.
- Built agentic workflows on DeepSeek (via OpenClaw), augmenting the model with proprietary operational knowledge so employees retrieve and input information in a fraction of the previous time, replacing a slow manual process.
- Drive product iterations against hours-saved and production-rate targets, translating operational pain points into measurable engineering outcomes.

SINGAPORE MANAGEMENT UNIVERSITY — Singapore

Feb 2026 – Apr 2026

Teaching Assistant

- Python Bootcamp: guided ~60 graduate students through environment setup and core Python programming.
- Statistics for Data Science: developed lab code and tutorial materials in Python for the professor in charge.

FREELANCE — Remote

Jul 2024 – Present

Freelance Software Engineer

- Partner directly with the client to scope evolving product requirements, independently delivering automation for customer tracking (~200 customers), financial reconciliation, and third-party CRM integrations for data ingestion and processing.
- Re-architected the solution across shifting requirements, moving from Java on GCP to Python scripts on AWS EC2 and finally a streamlined Google Apps Script deployment, right-sizing the stack for cost and maintainability.
- Cut recurring client workload from 2 days to 3 hours by automating manual operational and reconciliation processes.

TRAVELOKA — Indonesia

Jan 2022 – Jan 2025

Software Engineer

- Automated the end-to-end tax reporting workflow for business partners using CRON-triggered AWS Lambda functions, eliminating manual processing and saving ~500 man-hours per year across multiple accommodation partners.
- Designed and shipped a bulk-onboarding system in Java and Spring Boot with NoSQL on AWS (ECS, Lambda), enrolling 200+ hotel entities in a single batch process and cutting onboarding turnaround from months to roughly one week.
- Delivered a third-party hotel onboarding integration ahead of the planned production deployment despite evolving requirements, coordinating across multiple teams to contain scope spillover.

- Implemented PIN-based authentication with AWS Cognito for the business unit, strengthening security and enabling fine-grained user access control.
- Built CI/CD pipelines with GitHub Actions to containerise and publish AWS images for each microservice, deploying to AWS ECS instances and streamlining release cycles.
- Designed and developed OpenAPI-documented REST APIs on Spring Boot, standardising service contracts across the platform's microservices.

AI / ML PROJECTS

- **Cross-Category Recommender System:** Built an end-to-end recommendation pipeline on the McAuley-Lab Amazon Reviews 2023 dataset with RecBole and PyTorch; benchmarked sequential models (SASRecF, UniSRec) before selecting collaborative filtering (BPR, LightGCN) as final models, evaluated on NDCG@K and a harmonic-mean composite metric. Engineered cross-category enhancements with sentence-transformer embeddings to improve cold-interaction quality, prototyping custom RecBole architectures with Claude as a coding agent before manual fine-tuning.
- **Lawnet — Multi-Agent RAG for PDPA Compliance:** Engineered a multi-agent, retrieval-augmented generation system providing context-aware guidance on Singapore's Personal Data Protection Act, using autonomous agent workflows and vector-based retrieval powered by DeepSeek-V3.1 and Gemini 2.5 Pro, designed the prompt engineering strategy governing agent reasoning and tool use.
- **Crypto Illicit Transaction Classification:** Benchmarked Logistic Regression, Random Forest, XGBoost, Isolation Forest, and Graph Convolutional Network (GCN) classifiers on a small labelled dataset, lifting performance through active learning with targeted labelling.
- **End-to-End ML Development & Engineering:** Managed the full data lifecycle, feature engineering, data visualisation, and model optimization, for predictive modelling with Pandas and scikit-learn.

EDUCATION

SINGAPORE MANAGEMENT UNIVERSITY — Singapore

2025 – 2026

Master of IT in Business (Artificial Intelligence)

School of Computing and Information Systems

GPA: 3.72 / 4.0

- Relevant coursework: Recommender Systems, Agentic AI for Intelligent Autonomy, GenAI with LLMs, Applied Machine Learning, Deep Learning for Visual Recognition, Reinforcement Learning, AI & Uncertainty Reasoning, Algorithm Design & Implementation, Cybersecurity Technology & Applications, Query Processing & Optimisation (SQL), Statistics.

UNIVERSITY OF SYDNEY — Australia

2018 – 2022

Bachelor of Computing (Software Development)

School of Computer Science

- Relevant coursework: Data Structures & Algorithms, Systems Programming, Object-Oriented Programming, Machine Learning & Data Mining, Agile Software Development Practices.